Prerequisites

What a team should know before starting. The team as a whole should cover all three areas, and each member should be strong in at least one.

Role	Owns
RTL lead	ISA or state machine, datapath, pin map
Verification lead	Testbenches, reference models, formal and random testing
Flow / integration lead	Tiny Tapeout template, Crucible flow, timing, area, GDS

1. RTL-to-GDS

Design box: IHP 130nm SG13G2 via Tiny Tapeout · 6 × 4 tiles (~0.7 mm^2) · 26 pins (8 in, 8 out, 8 bidirectional, `clk`, `rst_n`) · single clock, active-low reset.

- **RTL:** synthesisable Verilog, FSMs, counters, shift registers, no latches, reset handling, input synchronizers, bidirectional and open-drain pins
- **Verification:** self-checking testbenches that fail on mismatch, waveform debug, reference models, basic Python for cocotb. Good to have: constrained-random, coverage, formal
- **Physical design:** know what each stage in Chip Lifecycle does and produces: lint, RTL, sim, verification, synthesis, DFT, floorplan, placement, STA, place and route, GDS, DRC/LVS. Be able to read a timing report and a DRC report

2. Protocols

References are in `specs/`.

Protocol	Status	Must know	Pins
UART	Mandatory	Frame format (start, data LSB first, parity, stop; 8N1), baud divisor, 16× oversampling, framing/parity/overrun errors	2
SPI	Mandatory	SCLK/MOSI/MISO/CS, modes 0 to 3 (CPOL/CPHA), controller vs peripheral, bit order, CS timing, full duplex	4
I2C	Mandatory	Open-drain SDA/SCL, START/STOP/repeated START, ACK/NACK, 7-bit address + R/W, clock stretching, arbitration, 100 and 400 kbit/s	2 bidir
Low-speed USB	Optional	1.5 Mbit/s differential D+/D−, NRZI, bit stuffing, SE0/EOP, packets and CRC (USB 2.0 ch. 7 and 8)	2 bidir
10 Mbit Ethernet	Optional	Manchester at 10 Mbit/s (20 MHz+ clock), frame format and CRC32, external analog/PHY (IEEE 802.3)	2 to 4

Take on the optional protocols only after UART, SPI and I2C pass.

3. Crucible knowledge

Crucible is the EDA platform by Ignytion IO. The whole flow, from RTL to GDS, runs through it, so every team member uses the same tools and versions. Learn it from these resources:

Resource	Link
Video tutorials	youtube.com/@Ignytionio
Digital design examples	github.com/CRUCIBLE-WORKBENCH/digital-design
Analog design examples	github.com/CRUCIBLE-WORKBENCH/analog-design
Whitepapers	whitepaper.ignytion.io

Questions: info@ignytion.io